MIL-B-2883D
17 October 1988
SUPERSEDING
MIL-B-2883C
20 Sep 1976

MILITARY SPECIFICATION

BOATSWAIN'S PIPE

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of nickel plated brass boatswain's pipe.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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SPECIFICATIONS

FEDERAL

* NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
QQ-N-290	- Nickel Plating (Electrodeposited)
UU-P-553	- Paper, Wrapping, Tissue
PPP-B-676	- Boxes, Setup
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

MILITARY

* MIL-P-15011	- Pallet, Material Handling, Wood, Post Construction, 4 Way Entry
MIL-B-17757	- Boxes, Shipping, Fiberboard (Modular Sizes)

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
* MIL-STD-147	- Palletized Unit Loads

(Copies of the Federal Information Processing Standards (FIPS) are available to Department of Defense activities from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pa 19120-5099. Others must request copies of FIPS from National Technical Information Service, 5285 Port Royal Road, Springfield, Va 22161-2171.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001)

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2.2 Non-Government. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the documents cited in the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also maybe available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Materials.

3.3.1 Copper base alloys.

3.3.1.1 Composition. The component parts of the boatswain's pipe shall be fabricated from any of the copper alloys specified in Table I, and shall conform to the applicable composition as indicated when tested as specified in 4.4.1. The copper alloy used shall be rolled polished, and shall be free from pits; scale (including red oxide); dents; nicks; cracks; scratches; segregations; and foreign inclusions that cannot be removed in later processing.

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Table I - Chemical composition of copper base alloys (percent)

Alloy	Copper	Nickel	Lead (max.)	Iron (max.)	Manganese	Zinc	Others (max.)
Yellow brass	64.0 to 68.5	-	0.15	0.05	-	remainder	0.15
Red brass	84.0 to 86.0	-	0.05	0.05	-	remainder	0.15
Nickel silver	63.0 to 66.5	16.5 to 19.5	0.10	0.25	0.50	remainder	-

3.3.2 Brazing alloys. The brazing alloys shall be of a quality suitable for the purpose intended, and shall be free of cadmium; lead; and other toxic ingredients.

3.4 Design. The boatswain's pipe shall consist of a one piece curved and tapered tube, with a hollow ball sound chamber at the tapered end and with either a folded-edge or a solid brass mouthpiece at the opposite end (see 6.4); a flat strip of blade positioned along the bottom rear half of the tube and with a split ring through a hole in the blade; and a continuous reinforcing strip along the blade edges and ball ending at the ball opening (see Figure 1).

3.4.1 Figure. Figure 1 is furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figure, the written specification shall govern.

3.5 Thickness. The minimum thickness of the applicable copper alloy used for the component parts shown below and prior to nickel plating, shall be as follows when tested as specified in 4.4.1.

Component	Requirement
Ball	0.016 inch
Tube	0.016 inch
Mouthpiece, folded edge when applicable	0.013 inch
Blade	0.032 inch
Reinforcement strip	0.016 inch
Split ring	0.057 inch

3.6 Construction. The component parts shall be formed from flat stock except for the mouthpiece when made from a solid brass part. The parts shall be neatly fitted together and brazed at the joints. The brazed connections shall be completely joined without excess brazing material, and all excess flux shall be removed when applicable.

3.6.1 Nickel plating. The assembled boatswain's pipe shall be nickel plated conforming to class 1, grade D, of QQ-N-290 (see 4.4.1).

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3.6.2 Identification markings. Unless otherwise specified (see 6.2), the boatswain's pipe shall be marked with the manufacturer's name, trademark, or other identification symbol on either the blade or centered on the reinforcement strip. The marking shall be 3/64 (±1/64) inch in height, and shall be permanent and legible.

3.7 Finish. The finished boatswain's pipe shall have a smooth finish, free from burrs; sharp edges; slivers; die marks; tool and grind marks; dents; cracks; and extraneous matter.

3.8 Finished dimensions.

3.8.1 Ball. The ball shall have an outside diameter of 3/4 (±1/32) inch, with a 1/4 (±1/32) inch diameter opening at the top.

3.8.2 Tube. The tube shall be 5 (±1/16) inch in length, tapering from 9/32 (±1/64) inch diameter at the mouthpiece to 5/32 (±1/64) inch diameter at the ball.

3.8.3 Mouthpiece. The folded edge or solid mouthpiece as applicable shall be 1/4 (±1/32) inch in length. The rolled or solid end shall have an outside diameter of 7/16 (±1/32) inch, with a 1/4 (±1/32) inch diameter opening. The folded edges of mouthpiece when applicable, shall extend 5/32 (±1/64) inch along the outside of mouthpiece.

3.8.4 Blade. The blade shall be 2-5/8 (±1/8) inches in length by 5/8 (±1/16) inch in width. The blade hole shall be 3/32 (±1/64) inch in diameter, and shall be approximately 3/16 inch from both the front and bottom edges of the blade when measured from the center of the hole.

3.8.5 Reinforcing strip. The reinforcing strip shall be 1/8 (±1/32) inch in width.

3.8.6 Split ring. The split ring shall have an outside diameter of 9/16 (±1/16) inch, after the ends are butted together and aligned.

3.8.7 Overall length. The overall length of the finished assembled boatswain's pipe shall be 5-1/2 (±1/8) inches.

3.9 Performance requirement. The finished boatswain's pipe shall emit an audible characteristic sound when tested as specified in 4.4.2.3.

3.10 Workmanship. The finished boatswain's pipes shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, appearance, performance and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be submitted and will be acceptable for the chemical composition of the brass (see 3.3.1.1) and for the thickness of the applicable component parts prior to nickel plating specified in 3.5. Testing for the thickness and adhesion of the nickel plating shall be as specified in 4.5.1 and 4.5.2 of QQ-N-290.

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4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.3. The applicable inspection levels and the acceptable quality levels, shall be as indicated in 4.4.2.4. The sample unit and the lot size shall be expressed in units of one boatswain's pipe.

4.4.2.1 Visual examination. The boatswain's pipes shall be examined for defects in color, design, material, workmanship, and marking in accordance with the list below:

Examine	Defect
Design	Not as specified
Components and Assembly	Any component part or operation omitted Any component part not as specified Any component part misplaced or not serving intended purpose Any component part defective, i.e., malformed, bent, cracked, split, twisted, or otherwise impaired Any area burned or reduced Brazed joints not complete, i,e., any opening at joint Brazed joints not clean and smooth, i.e., any flux or excess brazing alloy not removed Ends of split ring not closed or not in alignment
Finish	Plating omitted, not continuous, cut through, porous, blisters, pits, nodules or excessive edge build up, embedded foreign matter, peeling, flaking, or chipped
Identification Marking	Missing, incorrect, or illegible Not positioned as specified

4.4.2.2 Dimensional examination. The boatswain's pipes shall be examined for dimensional defects. Any dimension that is not within the established tolerance, shall be classified as a defect (see 3.8.1 thru 3.8.7).

4.4.2.3 Examination for performance. The boatswain's pipe shall emit an audible characteristic sound when blown by mouth. The absence of an audible characteristic sound shall constitute a defect (see 3.9).

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4.4.2.4 Inspection levels and acceptable quality levels (AQL). The inspection levels, expressed in defects per 100 units for visual and dimensional examination, shall be as follows:

	Inspection level	AQL
For defects applicable to 4.4.2.1	II	6.5
For defects applicable to 4.4.2.2	S-2	2.5
For defects applicable to 4.4.2.3	S-4	1.0

4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulging or distorted container.
Content	Number of boatswain's pipes per set-up box is more or less than specified. 1/ Number of set-up boxes per container is more or less than specified.

1/ For this defect, on set-up box shall be examined from each shipping container in the sample.

* 4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the acceptable quality level, expressed in terms of defects per hundred units, shall be 6.5 as specified in MIL-STD-105.

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Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be level A or C specified (see 6.2).

5.1.1 Level A. Each boatswain's pipe shall be completely wrapped in a sheet of tissue paper approximately 10 inches by 8 inches conforming to type II, class 1 or 2, of UU-P-553. Twelve (12) each boatswain's pipes shall be packaged in a setup paperboard box conforming to type I, variety 1, class A, style 4, of PPP-B-676. The boatswain's pipes shall be packaged flat and reversed end to end, six in length; one in width; and two in depth within the setup paperboard box. The outside dimensions of each setup paperboard box shall be 7-5/8 inches in length, 5-3/4 inches in width, and 1-3/8 inches in depth. Each setup paperboard box shall be securely closed with 2-inch minimum width gummed paper tape conforming to type III, class 2, grade B, of PPP-T-45, applied along the bottom and up each side a minimum of 3/4 inch.

5.1.2 Level C. Boatswain's pipes shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Two hundred forty (240) boatswain's pipes, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to class weather-resistant, grade V2s, size 1, of MIL-B-17757. Level A packages shall be packed flat, two in length, two in width, and five in depth within the shipping container.

5.2.2 Level B. Two hundred forty (240) boatswain's pipes, packaged as specified in 5.1, shall be packed in fiberboard shipping container assembled and closed conforming to class domestic, grade 275, size 1, of MIL-B-17757. Level A packages shall be packed flat, two in length; two in width; and five in depth within the shipping container.

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5.2.3 Level C. Boatswain's pipes, packaged as specified in 5.1, shall be packed in a manner to insure safe delivery at destination at the lowest transportation rate for such deliveries. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

* 5.4 Palletization. When specified (see 6.2), item, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means C, K and L, or O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, Type IV; Type V, class 1, size 2; or Type VIII, fabricated from wood groups I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contacting officer.

6. NOTES

6.1 Intended use. The boatswain's pipe is intended to be used by Navy and Coast Guard personnel for all purpose calls.

6.2 Acquisition data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
* b. When a first article inspection is required, (see 3.2), the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quantity, and testing and approval of the first article.
* c. Issue of DODISS to be cited in the solicitation, and if required, the specified issue of individual documents referenced (see 2.1.1 and 2.2).
d. Identification marking required if other than specified (see 3.6.2).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
* f. When palletization is required (see 5.4).

6.3 Samples. For access to samples, address the procuring activity issuing the invitation for bids.

6.4 Mouthpiece. The solid brass mouthpiece when applicable has been made satisfactorily using a screw machine or when die struck.

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* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.6 Subject term (key word) listing.

Brass, nickel plated
Mouthpiece, solid brass
Split ring

Custodian:
Navy - NU

Preparing Activity:
Navy - NU

Review Activity:
DLA - CT
Navy - CG

Project No. 8465-N009

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LEGEND OF COMPONENTS

A. BALL
B. TUBE
C. MOUTHPIECE
D. BLADE
E. REINFORCING STRIP
F. SPLIT RING

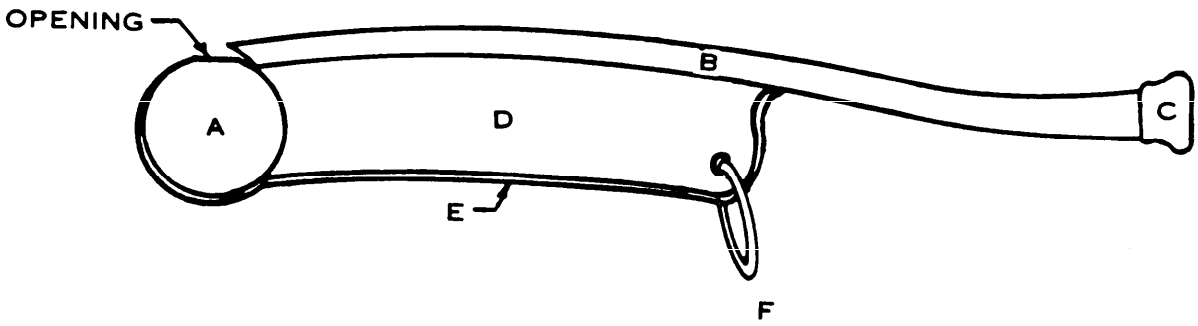


FIGURE 1. - BOATSWAIN'S PIPE

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-2883D	BOATSWAIN'S PIPE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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